1) $W(\text{care}) = np_c \qquad W(\text{desert}) = np_d + mnp_2$

Condition: $p_c = 2p_d, \quad p_c = 3p_2, \quad m = 0.5$

$W(\text{care}) = np_c$

$$W(\text{desert}) = \frac{np_c}{2} + 0.5\,\frac{np_c}{3} = \frac{2}{3}p_c$$

$\therefore W(\text{desert}) < W(\text{care})$

2) i) Marginal value theorem: $s_{opt}\,p'(s_{opt}) = p(s_{opt})$

$$p(s) = \frac{s - s_{min}}{k + s - s_{min}},$$

$$p'(s) = \frac{k + s - s_{min} - (s - s_{min})}{(k + s - s_{min})^2} = \frac{k}{(k + s - s_{min})^2}$$

$$\therefore \frac{s_{opt}\,k}{(k + s_{opt} - s_{min})^{\cancel{2}}} = \frac{s_{opt} - s_{min}}{\cancel{k + s - s_{min}}} \Rightarrow s_{opt}\,k = (s_{opt} - s_{min})(k + s_{opt} - s_{min})$$

$$\Rightarrow 0 = s_{opt}^2 - 2s_{min}s_{opt} - s_{min}(k - s_{min}) = 0$$

$$\Rightarrow s_{opt} = s_{min} + \sqrt{k s_{min}} \quad \square$$

ii) Let $p(s) = s^2/(1+s)^2$

$$\therefore p'(s) = \frac{2s(1+s)^2 - 2s^2(1+s)}{(1+s)^4} = \frac{2s\cancel{(1+s)}(1+\cancel{s}-\cancel{s})}{(1+s)^{\cancel{4}\,3}}$$

$$= \frac{2s}{(1+s)^3}$$

$$s_{opt} = \frac{P(s_{opt})}{P'(s_{opt})} = \frac{s^2/(1+s)^2}{2s/(1+s)^3} = \frac{1}{2}s(1+s)$$

$$= \frac{1}{2}s_{opt} + \frac{1}{2}s_{opt}^2$$

$$\Rightarrow 1 = \frac{1}{2} + \frac{1}{2}s_{opt} \Rightarrow \frac{1}{2} = \frac{1}{2}s_{opt} \Rightarrow s_{opt} = 1$$

3) i) $r = \left(\frac{M}{K(1-k^4)}\right)^{1/3}$

$M = \rho r^2 \left(1 - \frac{k^2}{2}\right) d$

$$m = \rho \left(\frac{M}{K(1-k^4)}\right)^{2/3} \left(1 - k/2\right) d$$

ii) Minimise m by varying k, thus take

$\frac{dm}{dk} = 0$. Let $g(k) = \rho d(1 - k/2)$

$$h(k) = \left(\frac{M}{K(1-k^4)}\right)^{2/3} = \left(\frac{M(1-k^4)^{-1}}{K}\right)^{2/3}$$

$g'(k) = -\rho d/2$

$$= \left(\frac{M}{K}\right)^{2/3} (1-k^4)^{-2/3}$$

$$h'(k) = \left(\frac{M}{K}\right)^{2/3} \left(-\frac{2}{3}(1-k^4)^{-5/3} \cdot -4k^3\right) = \left(\frac{M}{K}\right)^{4/3} \left(\frac{8}{3} k^3 (1-k^4)^{-5/3}\right)$$

Product rule:

$$\frac{dm}{dk} = \rho d \left(\frac{M}{k}\right)^{2/3} \left(\frac{k}{3(1-k^4)^{5/3}} \left(4k^2(2-k) - 3(1-k^4)\right)\right)$$

$$= 0$$

$$\Rightarrow \frac{k}{3(1-k^4)^{5/3}}\left(4k^2(2-k^2)-3(1-k^4)\right) = 0$$

$$\Rightarrow \frac{4k^{\cancel{3}\,2}(2-k^2)}{\cancel{3(1-k^4)^{8/3}}} = \frac{\cancel{3k}(1-k^4)}{\cancel{3(1-k^4)^{5/3}}}$$

$$\Rightarrow 4k^2(2-k^2)-3(1-k^4)=0$$

$$\Rightarrow k^4-8k^2+3=0$$

$\Rightarrow$ Quadratic formula: $k^2 = 4 \pm \sqrt{13}$

For k_{opt} between $(0, 1)$

$$\frac{dm}{dk} = pd\left(\frac{M}{k}\right)^{2/3}\sqrt{4}\left(\frac{k}{3(\sqrt{1+k^4})^{5/3}}\left(4k^2(2-k^2)-3(1-k^4)\right)\right)$$

$k_{opt} = \frac{1}{k}$

4) Specialist $W_s = \dfrac{e_f v_f}{1 + e_f t_h}$

Generalist: yield/time of fast prey $= e_f v_f$

yield/time of lean prey $= e_l v_l$

Total prey $= e_f + e_l$

$\therefore$ Average $\dfrac{\text{yield}}{\text{time}} = e_f v_f + e_l v_l$

$\therefore$ Time expenditure $= 1 + (e_l + e_f) t_h$

$$\therefore W_g = \frac{e_f v_f + e_l v_l}{1 + (e_l + e_f) t_h}$$

ii) $W_g > W_s \implies \dfrac{e_f v_f + e_l v_l}{1 + e_l t_h + e_f t_h} > \dfrac{e_f v_f}{1 + e_f t_h}$

$$\implies e_f v_f + e_l v_l > e_f v_f \left(\frac{1 + e_l t_h + e_f t_h}{1 + e_f t_h} \right)$$

$$> e_f v_f \left(1 + \frac{e_l t_h}{1 + e_f t_h}\right)$$

$$\Rightarrow \cancel{e_l} v_l > e_f v_f \left(\frac{\cancel{e_l} t_h}{1 + e_f t_h}\right)$$

$$\Rightarrow \frac{v_l}{t_h} > \frac{e_f v_f}{1 + e_f t_h}$$

iii) if the lean value per time to capture and consume is greater than average energy gain of the specialist, then the generalist does better.

$$e_f v_f + e_l v_l > e_f v_f \left(\frac{1 + e_l t_h + e_f t_h}{1 + e_f t_h}\right)$$

Lecture 2i

1) a)

		Opponent	
		S	G
Own	S	1/2	1/4
	G	3/4	1/8

b) $W(S,S) = 1/2$, $W(G,G) = 1/8$ (Pure strategies)

W